

Quaternions: An introduction of Abstract Structures

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Section 1

Quaternions, denoted by $\mathbb{H}$ in honor of Hamilton, are a number system that extends the complex numbers.

Note complex numbers operate in a 2D plane, quaternions represent a 4D mathematical space.

A quaternion is written as a linear combination of a scalar and three imaginary basis vectors:

$$q = a + bi + cj + dk$$

where $a, b, c, d \in \mathbb{R}$.

The fundamental structural rule of $\mathbb{H}$ is defined by Hamilton's famous equation:

$$i^2 = j^2 = k^2 = ijk = -1$$

From this, we derive the non-commutative multiplication table (e.g., $ij = k$, but $ji = -k$). This specific loss of commutativity is not a flaw; it is the exact algebraic property required to accurately encode rotations in 3D space.

Expanding to $\mathbb{O}$ and $\mathbb{S}$

The Quaternions are just one step in a sequential expansion called the **Cayley-Dickson construction**. This process doubles the dimension of an algebra, but at a severe structural cost.

Given an algebra A , we form a new algebra of pairs (a, b) with multiplication:

$$(a, b)(c, d) = (ac - d^*b, da + bc^*)$$

Tracing this pattern reveals a beautiful, yet decaying, structural hierarchy:

- Reals ($\mathbb{R}$, 1D): Fully ordered and well-behaved.
- Complex ($\mathbb{C}$, 2D): Loses ordering ($i > 0$ is meaningless).
- Quaternions ($\mathbb{H}$, 4D): Loses commutativity ($ab \neq ba$).
- Octonions ($\mathbb{O}$, 8D): Loses associativity ($(ab)c \neq a(bc)$).

If we push to 16 dimensions, we get the Sedenions ($\mathbb{S}$). Here, the structure breaks down further: we lose **alternativity**. $\mathbb{S}$ contains zero divisors (non-zero numbers that multiply to zero), meaning it is no longer a division algebra, halting the neat geometric applications found in $\mathbb{H}$ and $\mathbb{O}$.

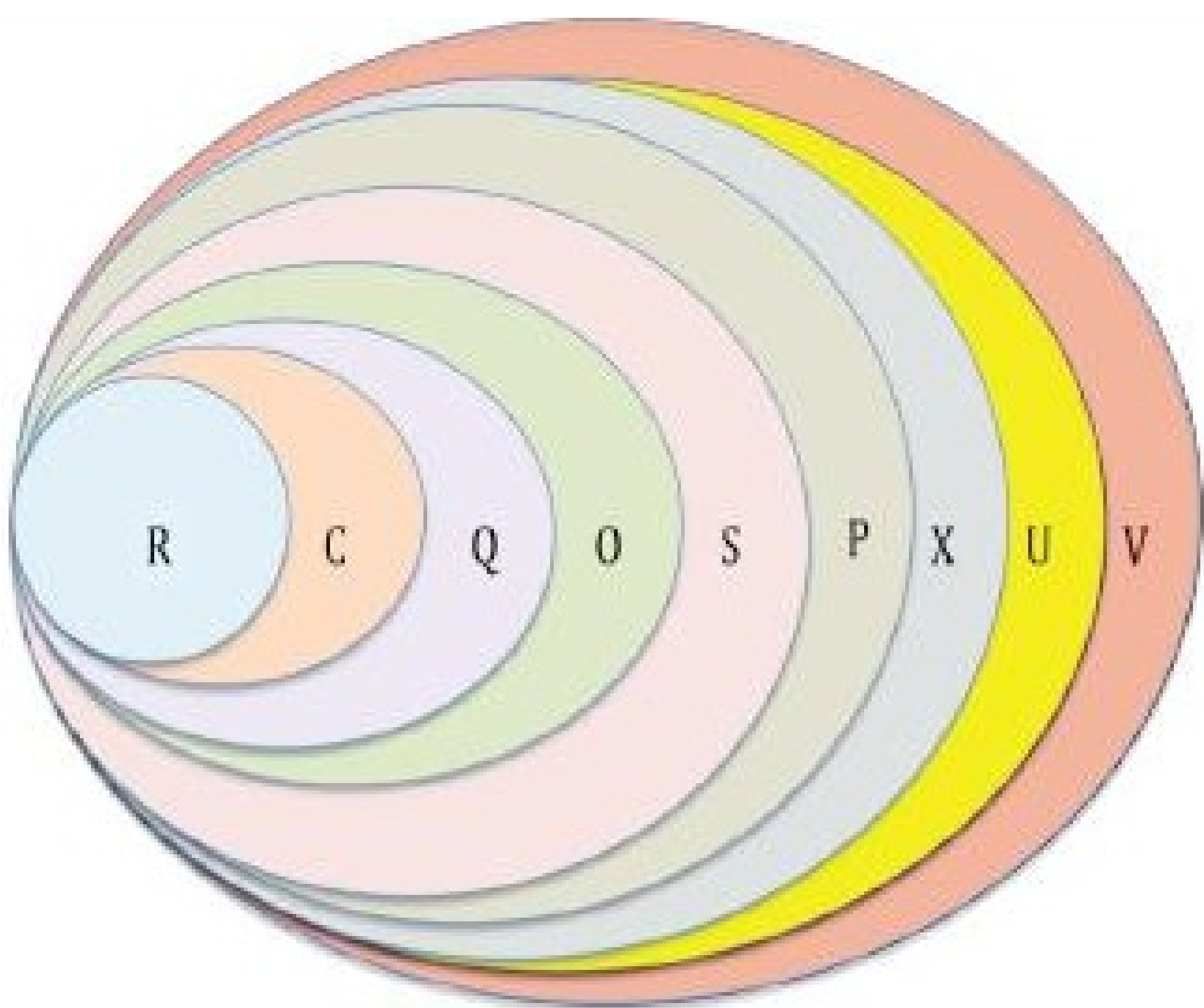


Figure 1: Enclosing Subsets of the first nine Cayley-Dickson Hypercomplex Algebras: Reals, Complex, Quaternion, Octonion, Sedenion, Pathion, Chingon, Routon, and Voudon

Geometry of $\mathbb{H}$ and $SO(3)$

We claim that the group of unit quaternions is intrinsically linked to $SO(3)$, the group of 3D rotations.

Let a 3D vector $v = (x, y, z)$ be represented as a **pure quaternion** (where the scalar $a = 0$):

$$v = xi + yj + zk$$

Now, let q be a unit quaternion ($\|q\| = 1$). We define an operation, the **sandwich product**, which transforms v :

$$v' = qvq^{-1}$$

Proof of rotation: First, we must ensure v' is still a pure quaternion (a valid 3D vector). The scalar part of qvq^{-1} vanishes, mapping 3D space to 3D space.

Second, does it preserve length? We evaluate the norm:

$$\|v'\| = \|qvq^{-1}\| = \|q\|\|v\|\|q^{-1}\|$$

Since $\|q\| = \|q^{-1}\| = 1$, we get $\|v'\| = \|v\|$. An isometry in 3D space that fixes the origin is a rotation.

If we construct q using an angle θ and a unit rotation axis $u = u_xi + u_yj + u_zk$:

$$q = \cos\left(\frac{\theta}{2}\right) + u \sin\left(\frac{\theta}{2}\right)$$

Applying the sandwich product rotates v by exactly θ around u .

The Double Cover: Notice that q and $-q$ produce the exact same rotation, because $(-q)v(-q)^{-1} = qvq^{-1}$. Therefore, $SO(3)$ is not strictly isomorphic to the unit quaternions, $Sp(1)$. Instead, there is a 2-to-1 surjective homomorphism $Sp(1) \rightarrow SO(3)$. $Sp(1)$ acts as a “double cover” of 3D rotational space.

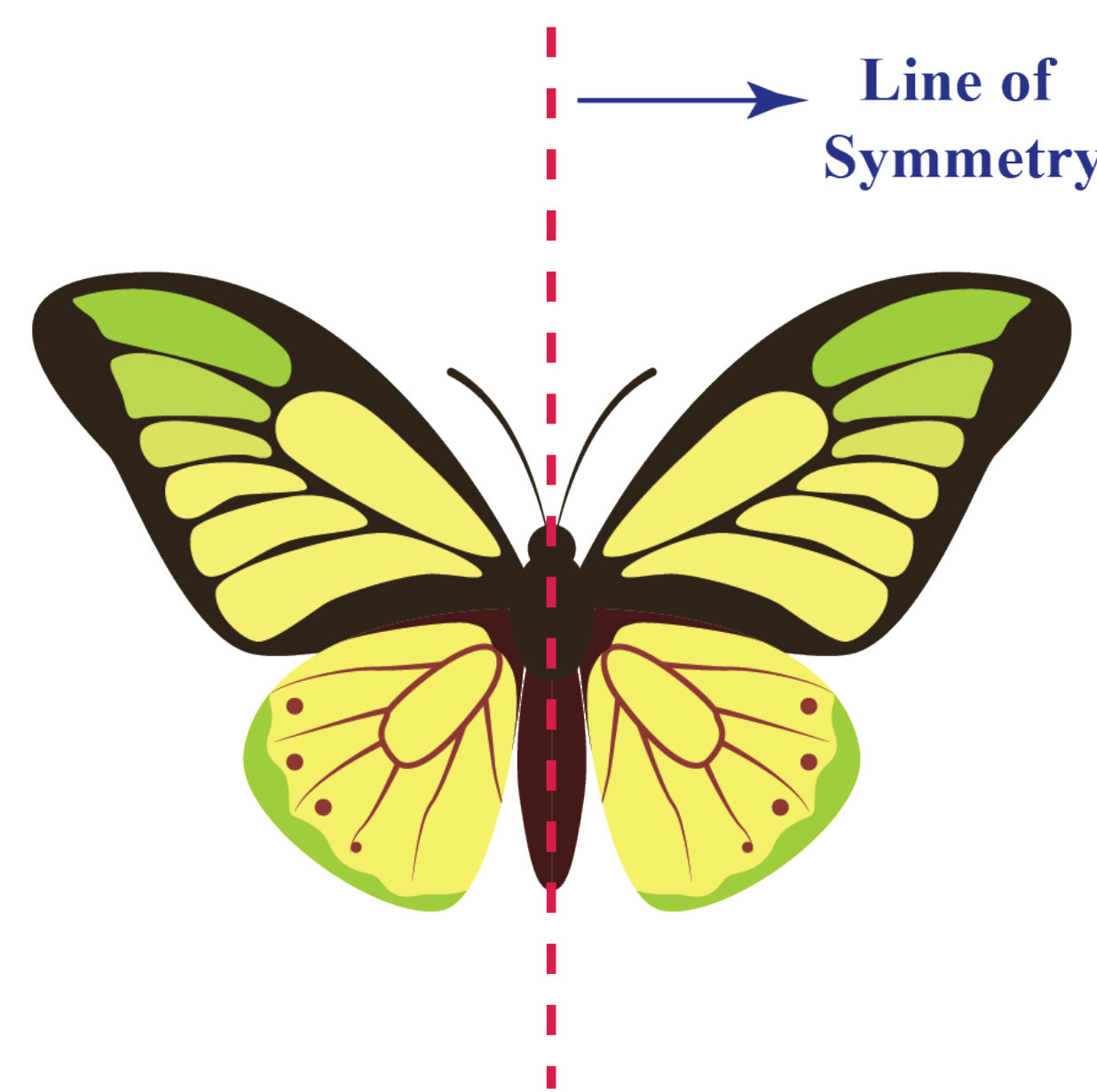


Figure 2:

Acknowledgements

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Applications and Symmetries

The rigid algebraic structure of these higher-dimensional spaces leads to profound applications across physics and computer science.

- Avoiding Gimbal Lock ($\mathbb{H}$): Traditional Euler angles (pitch, yaw, roll) suffer from gimbal lock—a loss of a degree of freedom when axes align. Because quaternions operate in a 4D hypersphere, they mathematically bypass this singularity.
- Smooth Interpolation ($\mathbb{H}$): Quaternions allow for SLERP (Spherical Linear Interpolation). Interpolating between two unit quaternions traces the shortest path on the S^3 sphere, creating perfectly uniform rotational animations in robotics and 3D graphics.
- Exceptional Symmetries ($\mathbb{O}$): While Sedenions are too chaotic, Octonions possess just enough structure to map the symmetries of string theory. The automorphisms of $\mathbb{O}$ form the exceptional Lie group G_2 , a vital component in 11-dimensional M-theory.

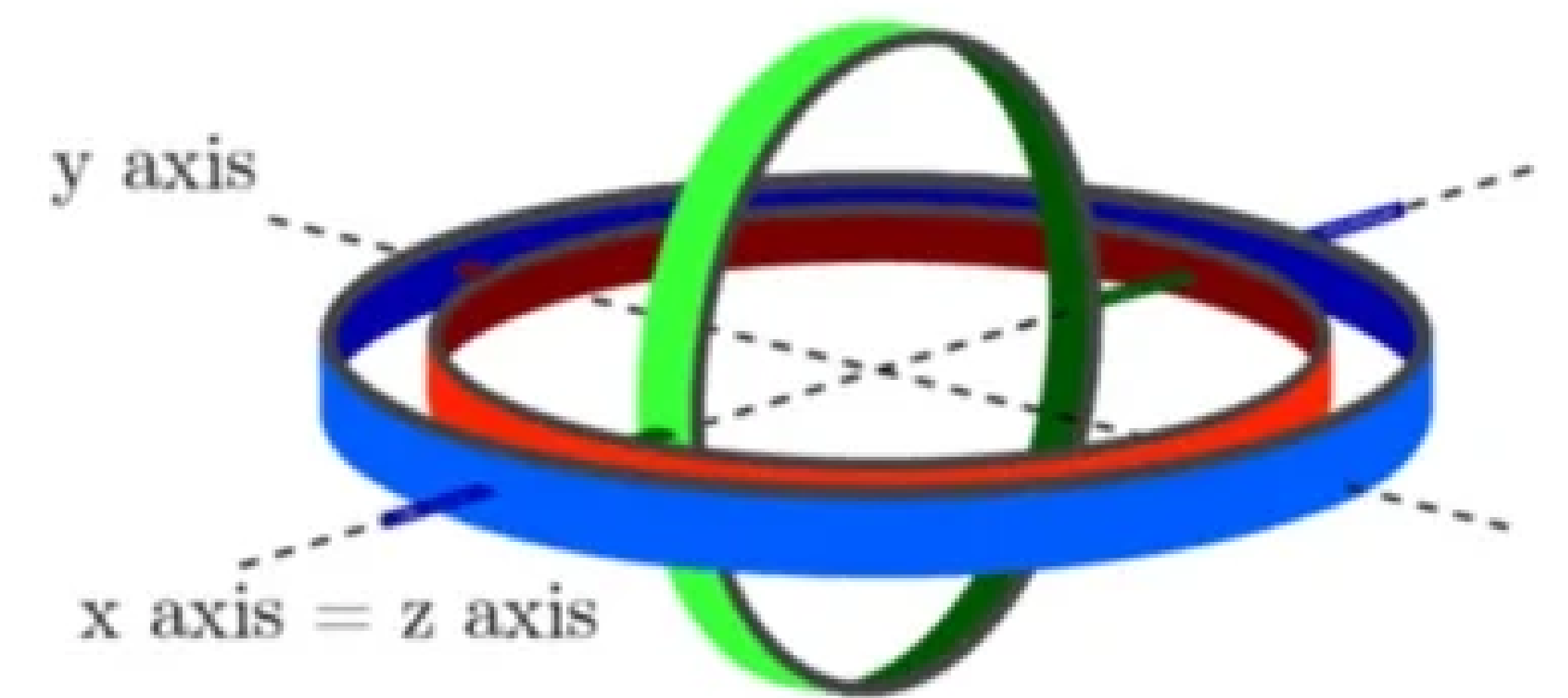


Figure 3:

Synopsis

- Algebra dictates Geometry: The rules of multiplication in a number system directly determine the spatial transformations it can represent.
- The Dimension Tax: Expanding dimensions via Cayley-Dickson grants more descriptive power but exacts a strict toll on algebraic stability (losing commutativity, then associativity).
- The Magic of 4D: $\mathbb{H}$ strikes the perfect balance—sacrificing commutativity is exactly what is needed to capture the non-commutative nature of 3D rotations (rotating around X then Y is not the same as Y then X).

References

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